5.3.4

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Question

Find the product of the matrices $\begin{pmatrix}1 & 2 & -3\\ 2 & 3 & 2\\ 3 & -3 & -4\end{pmatrix}\begin{pmatrix}-6 & 17 & 13\\ 14 & 5 & -8\\ -15 & 9 & -1\end{pmatrix}$ and hence solve the system of linear equations: $x + 2y - 3z = -4$
$2x + 3y + 2z = 2$
$3x - 3y - 4z = 11$

Solution

The given matrices are

$$\mathbf{A} = \begin{pmatrix}1 & 2 & -3\\ 2 & 3 & 2\\ -3 & 2 & -4\end{pmatrix}, \qquad \mathbf{B} = \begin{pmatrix}-6 & 14 & -15\\ 17 & 5 & 9\\ 13 & -8 & -1\end{pmatrix} \tag{1}$$

The product of the matrices is computed as

$$\mathbf{AB} = \begin{pmatrix}67 & 0 & 0\\ 0 & 67 & 0\\ 0 & 0 & 67\end{pmatrix} = 67\mathbf{I}_3 \tag{2}$$

The given system of linear equations can be written in matrix form as

$$\mathbf{AX} = \mathbf{C}, \tag{3}$$

$$\mathbf{C} = \begin{pmatrix}-4\\ 2\\ 1\end{pmatrix} \tag{4}$$

Since $\mathbf{AB} = 67\mathbf{I}_3$, the inverse of $\mathbf{A}$ exists and is given by

$$\mathbf{A}^{-1} = \frac{1}{67}\mathbf{B} \tag{5}$$

Multiplying both sides of $A\mathbf{X} = \mathbf{C}$ by $\mathbf{A}^{-1}$, we obtain

$$\mathbf{X} = \mathbf{A}^{-1}\mathbf{C} = \frac{1}{67}\mathbf{BC} \tag{6}$$

$$\mathbf{BC} = \begin{pmatrix} 201 \\ -134 \\ 67 \end{pmatrix} \tag{7}$$

Therefore, the solution of the system is

$$\mathbf{X} = \frac{1}{67}\begin{pmatrix} 201 \\ -134 \\ 67 \end{pmatrix} = \begin{pmatrix} 3 \\ -2 \\ 1 \end{pmatrix} \tag{8}$$

$$\tag{9}$$